Activity 1

Stand before the mirror in your dressing table or the mirror fixed in a steel almirah. Do you see your whole body? To see your entire body in a mirror, the mirror should be atleast half of your height. Height of the mirror= Your height/2.

6.3 Curved Mirrors

We studied about laws of reflection. These laws are applicable to all types of reflecting surfaces including curved surfaces. In your earlier classes, you have studied that there are many types of curved mirrors, such as spherical and parabolic mirrors. The most commonly used type of curved mirror is spherical mirror.

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6.3.1 Spherical mirrors

In curved mirrors, the reflecting surface can be considered to form a part of the surface of a sphere. Such mirrors whose reflecting surfaces are spherical are called spherical mirrors.

In some spherical mirrors the reflecting surface is curved inwards, that is, it faces towards the centre of the sphere. They are called concave mirrors. In some other mirrors, the reflecting surface is curved outward. They are called convex mirror.

6.3.2 Image formed by spherical mirrors

Activity 2

Hold a concave mirror in your hand (or place it in a stand). Direct its reflecting surface towards the sun. Direct the light reflected by the mirror onto a sheet of paper held not very far from the mirror. Move the sheet of paper back and forth gradually until you find a bright, sharp spot of light on the paper. Position the mirror and the paper at the same location for few moments. What do you observe? Why does the paper catches fire?

We saw that the parallel rays of sun light could be focused at a point using a concave mirror. Now let us place a lighted candle and a white screen in front of the concave mirror. Adjust the position of the screen. Move the screen front and back. Note the size of the image and its shape. You can see a small and inverted image.

Slowly bring the candle closer to the mirror. What do you observe? As you bring the object closer to the mirror the image becomes bigger. Try to locate the image when you bring the candle very close to the mirror. Are you able to see an image on the screen? Now look inside the mirror. What do you see? An erect magnified image of the candle is seen. In some positions of the object an image is obtained on the screen. However, at some positions of the object no image is obtained. It is clear that the behaviour of the concave mirror is much more complicated than the plane mirror.

However, with the use of geometrical technique we can simplify and understand the behaviour of the image formed by a concave mirror. In the case of plane mirror, we used only two rays to understand how to get full image of a person. But, for understanding the nature of image formed by a spherical mirrors we need to look at four specific rules.

6.3.3 Rules for the construction of image

To find the position and nature of the image formed by a spherical mirror, we need to know the following rules.

Rule 1: A ray passing through the centre of curvature is reflected back along its own path (Figure 6.4).

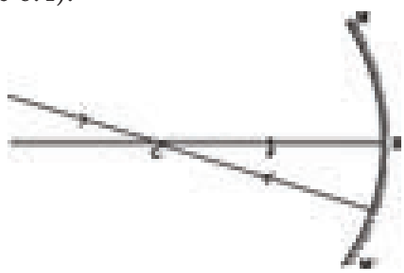


Figure 6.4 Ray passing through centre of curvature

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